

Lab 1 – Redflag

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CS411W

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September 27, 2026

Final Version

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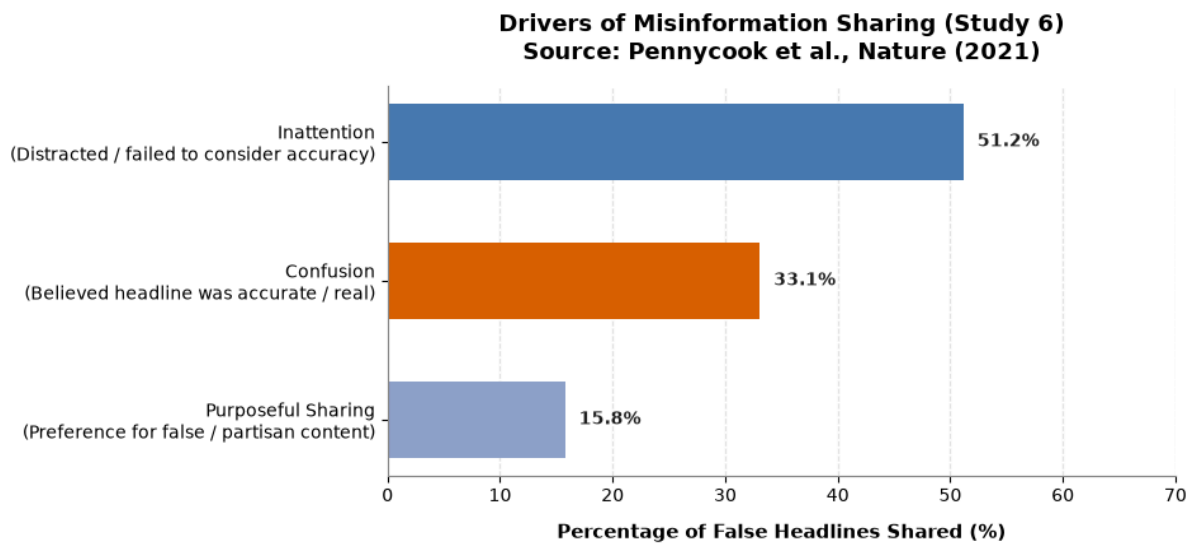
1 Introduction

Misinformation is spread across many social media platforms across the internet making it difficult for users to find reliable and trustworthy content. For users that typically browse through social media or news sources, there are no apparent signs that the information they come across is biased or misinformation. This leads to them sharing this misinformation as about 33.1% of false headlines are shared from users believing the information was real (see Figure 1). One reason for why users are unlikely to look further into the information they see is real is because it takes away from their experience of browsing. Users browsing social media typically are not on the lookout for misinformation, and are occupied with the experience of taking in media and moving on (Ordway, 2021). Existing solutions assist with identifying the information; however, all solutions detract from the original browsing experience and are seen as obtrusive. Another issue with the spread of misinformation is that the users, if they decide to try to look further into misinformation, will be spending more time on just finding good sources of information. It can be hard to judge and differentiate what other sources may also be spreading misinformation as well or if they are biased. There are no real time tools in the current ecosystem of fact checking directly within the browsing experience. A proper solution to this problem would provide a non-intrusive way of pulling facts in without being obtrusive within a browser. RedFlag addresses this gap by informing users with a balanced range of sources. Rather than explicitly pointing out misinformation to the user and popping up with sources of why it may be wrong, it is subtle in its implementation of letting the user know if they may want to look further into the information they see. RedFlag also solves the issue of the user needing to find reliable sources for misinformation. Rather than having the user decide on which sources are trustworthy

or what bias they may have, RedFlag curates these sources and balances them all out so no one source is over-represented.

Figure 1

Drivers of Misinformation Sharing Online



Note. From "Shifting attention to accuracy can reduce misinformation online," by G. Pennycook, Z. Epstein, M. Mosleh, A. A. Arechar, D. Eckles, and D. G. Rand, 2021, Nature, 592(7855), p. 593 (<https://doi.org/10.1038/s41586-021-03344-2>). Copyright 2021 by Springer Nature Limited.

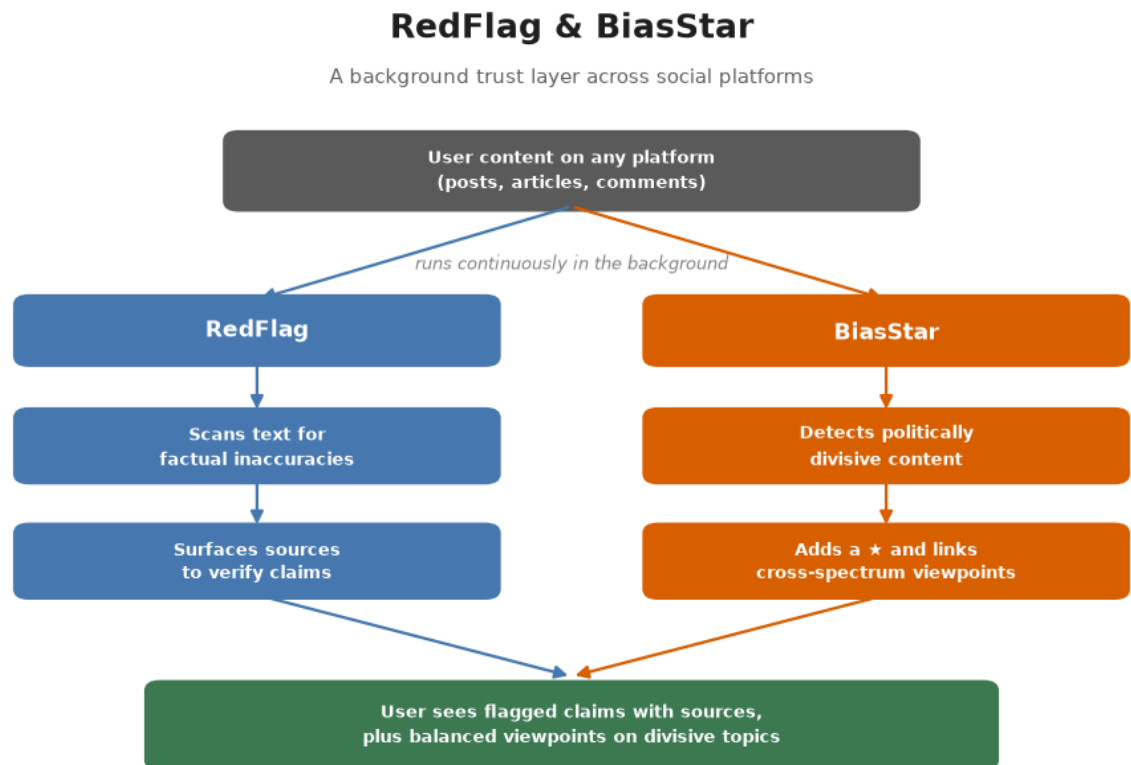
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2 Product Description

RedFlag is a web extension that provides real-time fact checking and awareness of content with political bias. It scans web content in the background to find any disinformation and provide accurate sources to bring awareness to the user. The BiasStar works alongside RedFlag's highlighting to note any politically divisive content and provide several viewpoints from different political perspectives.

Figure 2

High level concept diagram



2.1 Key Product Features and Capabilities

RedFlag offers real-time detection of misinformation while a user is scrolling through any social media. This provides a way of automatically pulling in sources and the ability to point out misinformation to the user without further manual input. This will allow the user to continue using social media the way they have before, without having to adjust to a different way of using social media. Users will be able to easily transition to using RedFlag without noticing it immediately unless they wish to.

RedFlag highlights questionable information subtly on the user's screen. The user will not be forced to see a popup and be interrupted, the highlights are minimal, but visible, for the user to be able to look further into the misinformation if they wish. This feature allows the user to continue on with their social media routine without interrupting their browsing session.

Another section of RedFlag includes BiasStar, which is displayed regarding any content that may be political. BiasStar will identify political content and provide multiple viewpoint sources across the political spectrum. The purpose of this is to allow the user to evaluate political content from every perspective, without any forced skewing to any one side. This will help tackle the issue of how social media may reinforce biases and the creation of echo chambers.

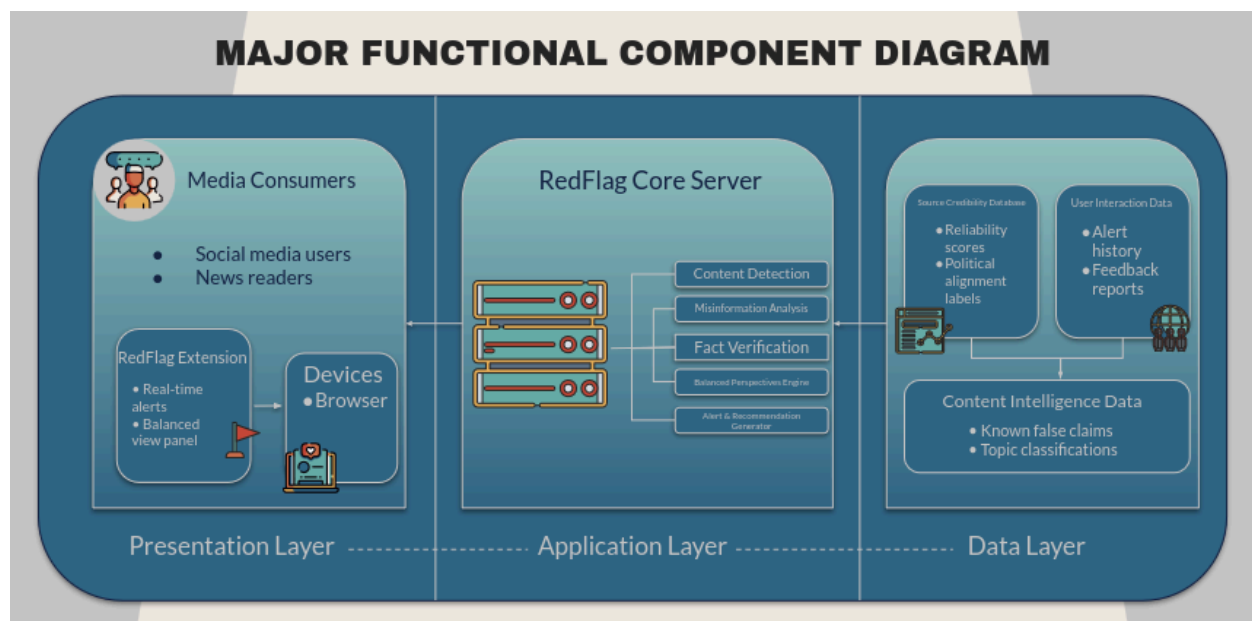
With all of the subtle highlights and BiasStar, the user is able to highlight the flagged information to see a more detailed overview. The detailed overview includes the further description of why content may be misinformation and any political information included with BiasStar. The detailed overviews will allow users to make a more informed decision, while still letting the user choose for themselves how to think on a topic rather than dictating what the user should think. This overall allows users to understand credibility behind any content and think further on it.

2.2 Major Components (Hardware/Software)

For users to run RedFlag, they will require a computer with a browser that supports the RedFlag extension. The main backend software of RedFlag that runs all the analysis will be run remotely with several server types. This includes a database server to hold the gathered data, a web server for administration, and general scripting server for querying Application Programming Interfaces (API).

Figure 3

Major Functional Component Diagram (MFCD)



RedFlag will comprise a presentation layer, application layer, and data layer. The presentation layer will be the acting front end where users will interact with a browser extension. This will display real time alerts and a view panel for users. It will be built with JavaScript for browser extensions. The application layer will act as the back end, running all the analysis and checks in the background for the user. This includes content detection, misinformation analysis, fact verification, alerts, the balanced perspectives engine, and the alert & recommendation

generator. It will be built with Python, more specifically a mix of Flask and Django, for the information datasets. All of the data will be held in the data layer, within a database. The database chosen for RedFlag is PostgreSQL. The stored data will include source credibility, user interaction data, and content intelligence data. The source credibility will include the reliability scores for sources and political alignments. The content intelligence data will include false claims and topic classifications.

For external interfaces and APIs, these include a fact checking API, news API, and LLM API. The fact checking API will simply compare questionable information against RedFlag's database. The news API will web scrape data from news and categorize by topic in the database. The Large Language Model (LLM) API will be for parsing the databases, summarizing data, and increasing fact-checking speed.

<Note: This section is intentionally left blank.>

3 Identification of Case Study

RedFlag is being developed for users who wish to be more informed of social media content they are consuming but find it distracting to look further into the specific details. It will help alleviate the issue of being intrusive to the user's experience by adding subtle highlights to content within the user's browser. RedFlag can further allow users to stay informed on political viewpoints of a topic without influencing the user's bias. This enables more people to become aware of the content they consume and provide an easier way to be informed of misinformation. Current fact-checking methods require significant effort from the user.

The primary study group for this is students participating in a college course on journalism. RedFlag would enable these students to quickly identify misinformation in real time while also finding reliable sources to enhance their productivity. Instructors can work alongside the students to help guide them through the proper usage of RedFlag and help contribute to the verification of RedFlag sources. In this case study environment, the students will be able to use the features of RedFlag for finding sources for their course work. While browsing, misinformation will be highlighted for the user to see, with a popup to allow further inspection of the content. Within the popup they will be able to review the sources and help add further verification to these claims.

For the journalism student, a day in the life of using RedFlag would start in the morning for John Doe. Around 10:00 AM, John checks his favorite news site for the morning news. RedFlag will add highlights to any misinformation detected while he scrolls through his morning news. Around 1:00 PM at his class, he decides that needs to conduct research on a particular topic about biology. RedFlag assists in flagging any questionable content that he comes across for reviews of the scholarly articles he looks at. At the end of his day, around 7:00 PM, he

decides to relax by scrolling through social media. RedFlag comes in to detect misinformation alongside BiasStar to help provide several viewpoints on political content.

Looking beyond individual users, this can be incorporated into schools to promote better awareness of misinformation online. It can become a helpful utility in a writer's toolkit to quickly grasp articles and understand online content for their own articles or publications. Researchers can use it to assist in identifying key information in articles with less worry of if the information is misinformation or misleading. RedFlag has the potential to help any person stay properly informed, whether it be current or past events.

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4 Glossary

Confirmation Bias: The tendency to interpret new information and evidence as confirmation of one's existing beliefs or theories, often while ignoring or discounting opposing evidence.

Credibility Score: A score given to a claim that is found online rating the credibility of the claim, or the likelihood that the claim is correct.

Disinformation: False or inaccurate information that is shared with the intention to mislead others.

End-user: The person who ultimately uses or is intended to use a computer system or software application.

Friction: Something that causes an application to not be seamless while working.

LLM (Large Language Model): A Large Language Model (also known as a ChatBot) such as ChatGPT, Gemini, or Claude.

Misinformation: Provably false or inaccurate information.

Troll: A person who posts inflammatory, insincere, creating discord, or off-topic messages in an online community.

5 References

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